UNIVERSITÀ DEGLI STUDI DI MILANO
DIPARTIMENTO DI MEDICINA VETERINARIA
E SCIENZE ANIMALI

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Herd, buffalo, and quarter specific prevalence and risk factors of subclinical mastitis in water buffaloes of Bangladesh

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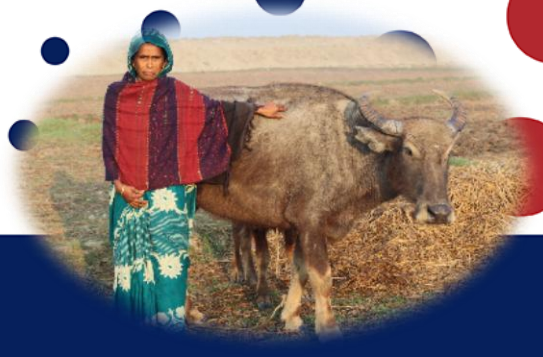


Background

Buffalo in Bangladesh

- 1.5 million buffalo (DLS, 2020)
- Concentrated in the coastal areas
- Mainly indigenous breeds and poor milk producers
- Scattered cross breeds are available in few intensive farms in some areas
- Milk production

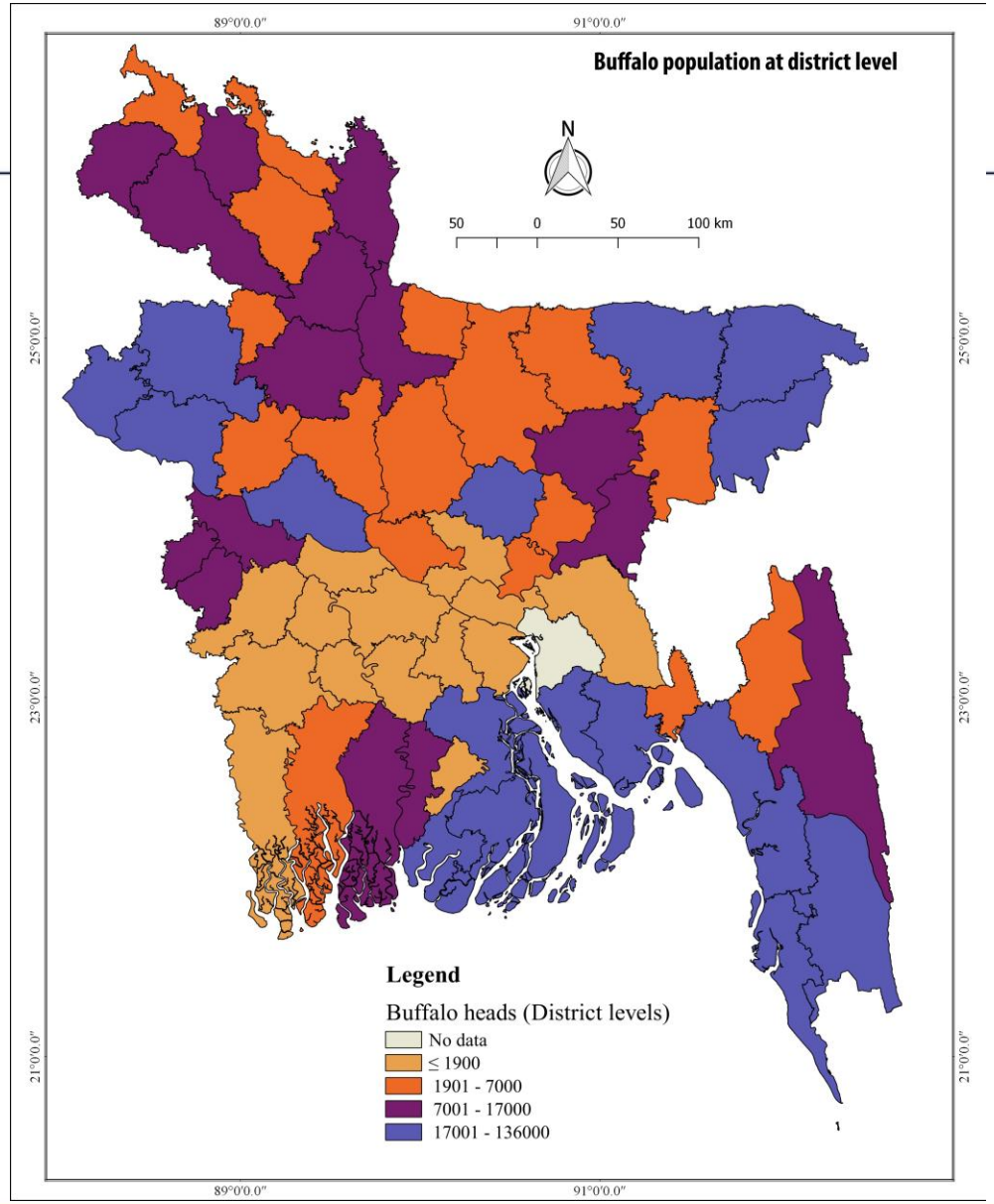
Indigenous: 0.5-3 L/ Day; Crossbreed: 3-5 L/ Day

Indigenous

Murrah

Jafrabadi

Nili-Ravi



Buffalo rearing systems

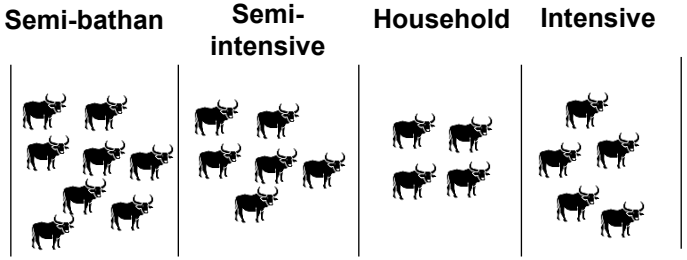
Island

Inland

Bathan → Killa → Island

Some bathans together form a killa

In one island there are several killa



Semi-bathans and intensive farms has more numbers of animals than the semi-intensive and households

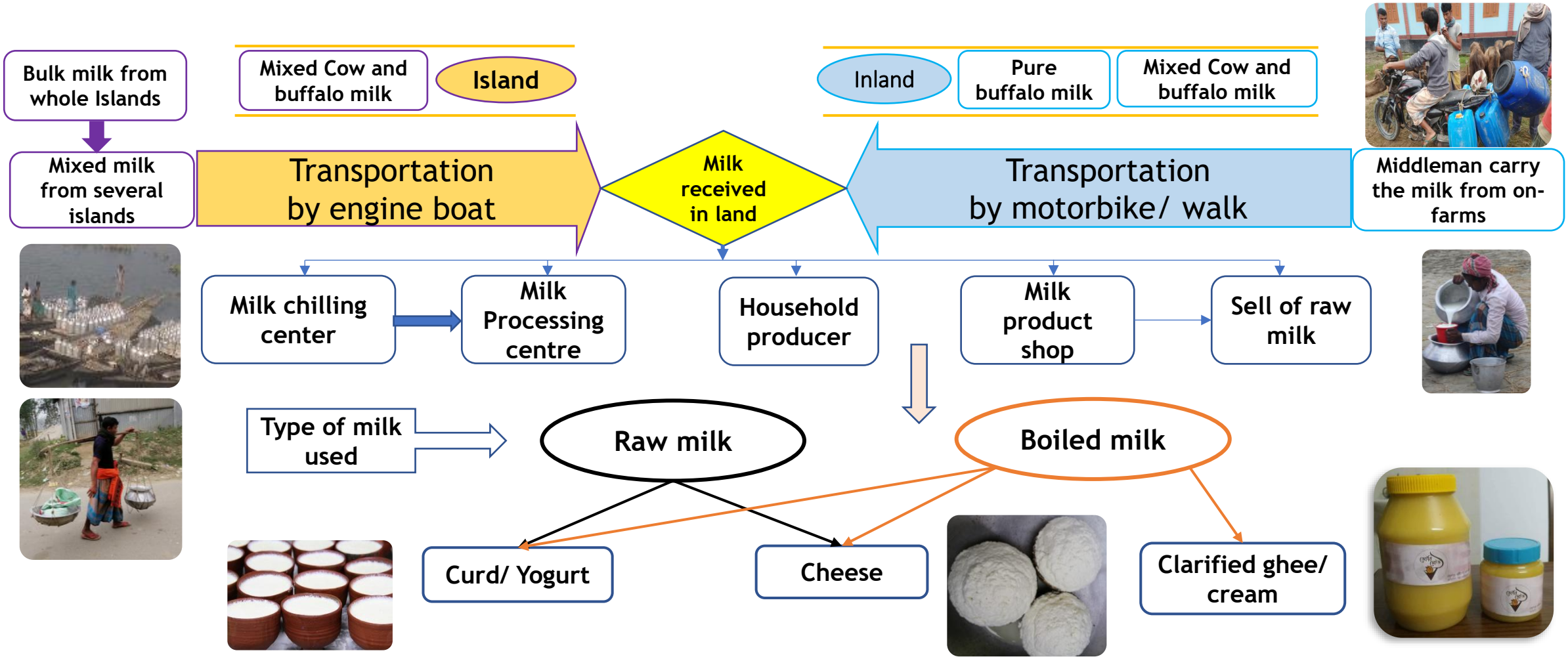
In Bathans

- Each farmer have 20-1000 buffalo, cows and sheep in a "bathan"
- Buffalo graze whole day and not provided any concentrate supplements
- There is no shed for the buffalo, they stay in open island

In Inlands

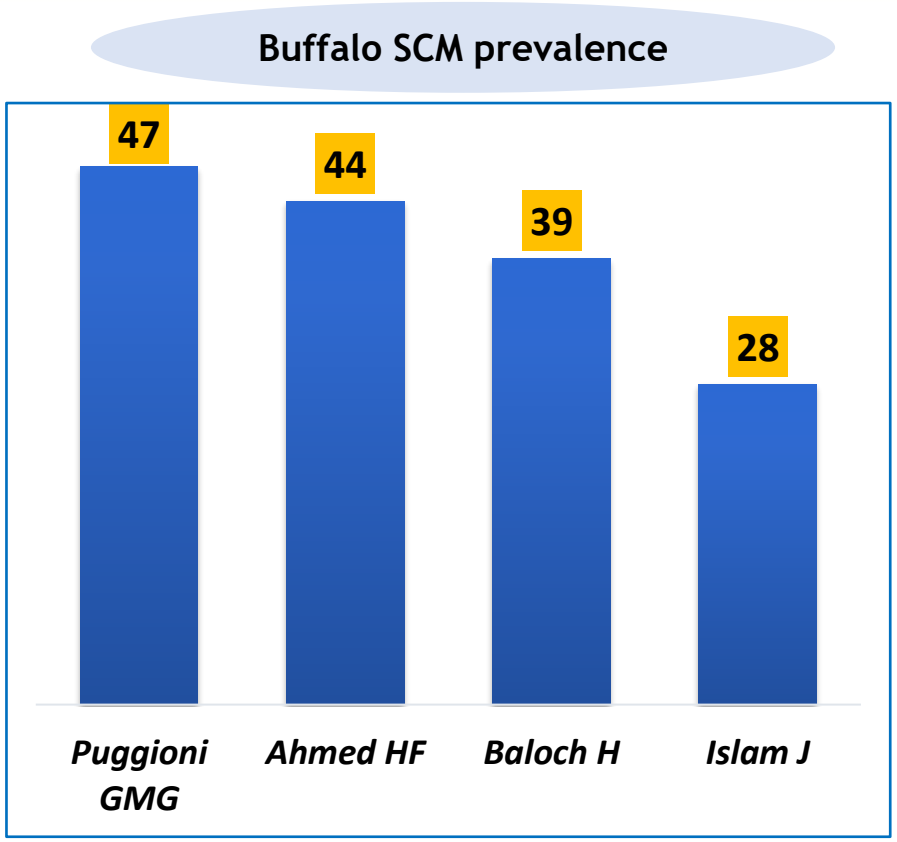
- Semi-bathan: Buffalo come back from bathan and stay in inlands for certain period (3-6 months)
- Semi-intensive: Buffalo stay at sheds, go to graze and come back everyday
- Household: Buffalo stay at shed or under tree and allowed to graze
- Intensive: Buffalo tied at sheds, stall fed and are never allowed to graze

Buffalo milk chain in Bangladesh



Mastitis is a major challenge for dairy buffalo

- Animal welfare is ignored [1]
- Fluctuation in surrounding temperature and humidity [2]
- Insufficient milking hygiene [1]
- Feed scarcity [1]



Missing the broader picture of SCM prevalence/ IMI pathogens in Bangladesh

1. Alam al., (2020). Animal-Based Welfare Assessment of Cattle and Water Buffalo in Bangladeshi Slaughterhouses. Journal of Applied Animal Welfare Science, 23(2), 219-230.
2. Ahmad, and Tariq. (2010). Heat stress management in water buffaloes: A Review. Revista Veterinaria. 21. 301-314.

Risk of microbial contamination in the buffalo milk chain in Bangladesh

Buffalo farms are located remotely

Poor communication and transportation facility

Insufficient milking hygiene

Inadequate milk transportation precaution

Longer transportation time

- Little is known about buffalo milk chain in Bangladesh

Udder Health Bangladesh

UHB
সুস্থ ওলান, নিরাপদ দুধ

Climate Change Mitigation by a sustainable water buffalo dairy chain in Bangladesh

SVA & SLU, Sweden

UU & WUR, The Netherlands

UNIMI & CReNBuf, Italy

CVASU & SAU, Bangladesh

2019-2023

Funded by

Swedish Research Council

Objectives

- Evaluate the buffalo milk chain
- Model the effect of transition from cows to buffalo milk chain
- Design a water buffalo milk chain

Materials and Methods

Sampling

Study period

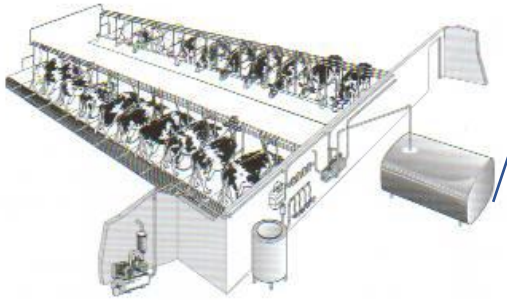
September-October 2019

February, 2020-March, 2021

- ❑Lactating buffalo
- ❑Functional quarters
- ❑Selected buffalo pockets

(11 districts) in Bangladesh

Source population

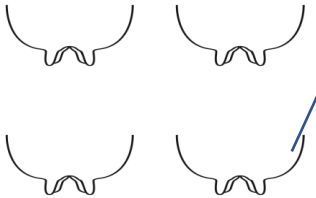


Farm

248 farms

Buffalo Cow

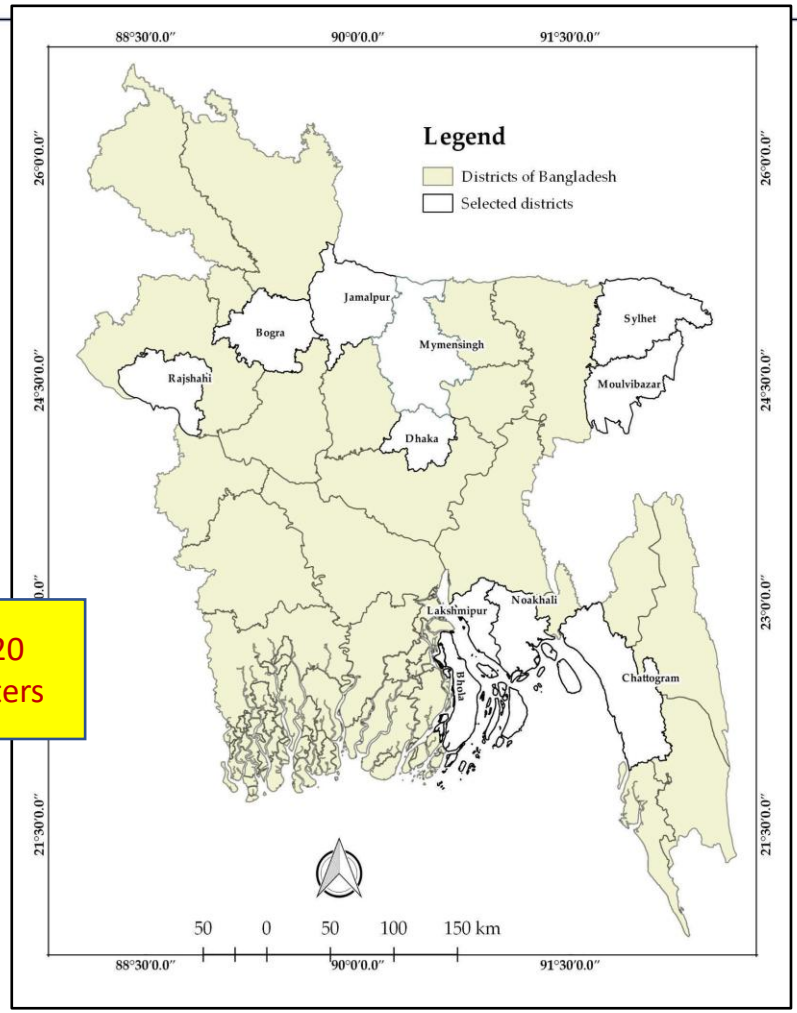
880 lactating buffalo



Quarter

3,520 quarters

Epi unit



Detection of subclinical mastitis in buffalo using California mastitis test

CMT of milk from **each functional quarter** with no clinical signs of mastitis

Scoring for milk for SCM detection
(Score 1-5)

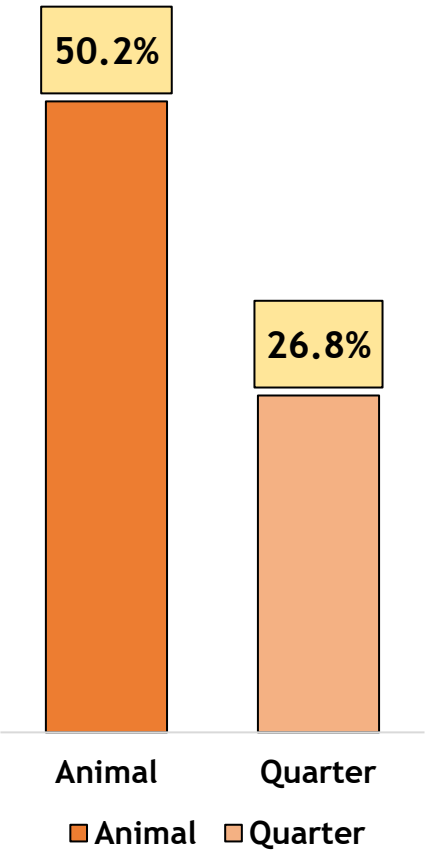
Aseptically collection and storage of samples
-Quarter samples and **Bulk milk somatic cell count (BMSCC)** using DeLavel Somatic cell counter

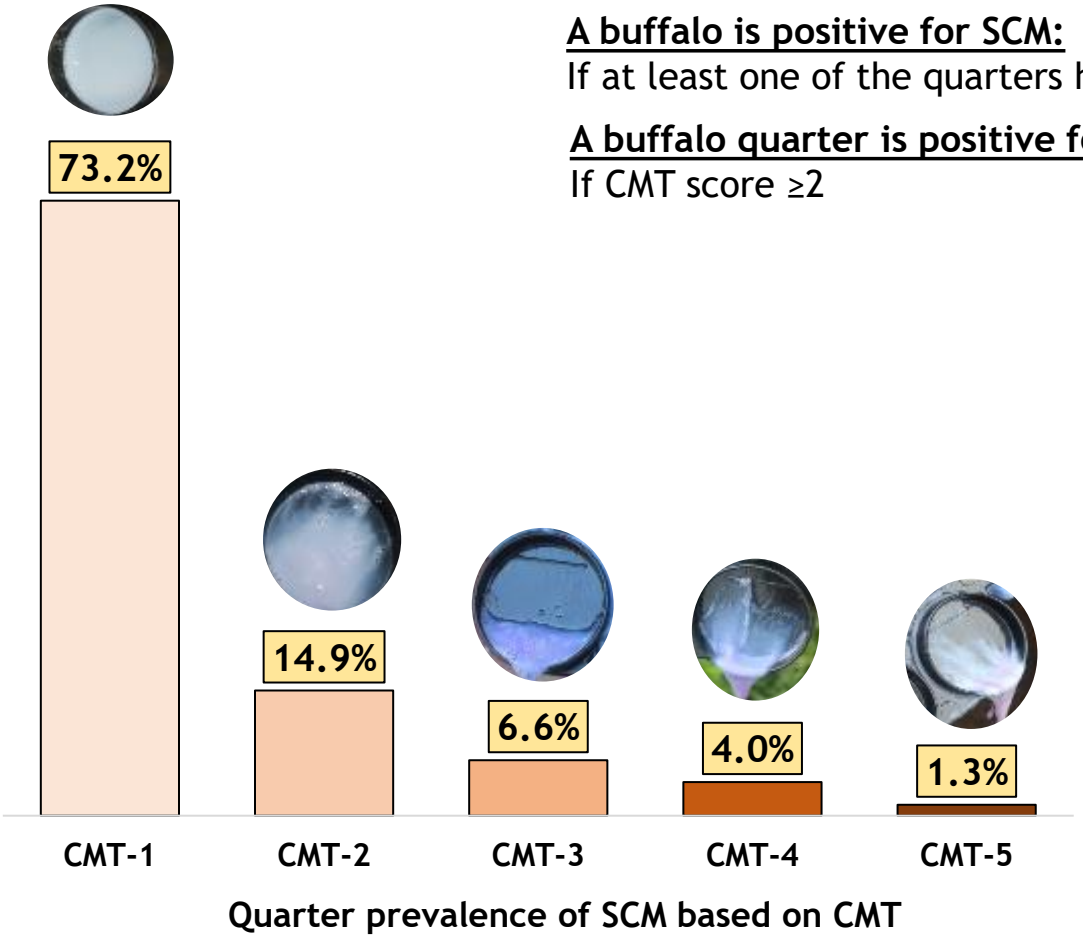
Transportation of samples to CVASU lab for bacteriological culture

MALDI TOF MS at Lab of Microbiology, Dept of Veterinary and Animal Science, UNIMI

Results

Prevalence of buffalo subclinical mastitis





A buffalo is positive for SCM:
If at least one of the quarters has CMT ≥2

A buffalo quarter is positive for SCM:
If CMT score ≥2

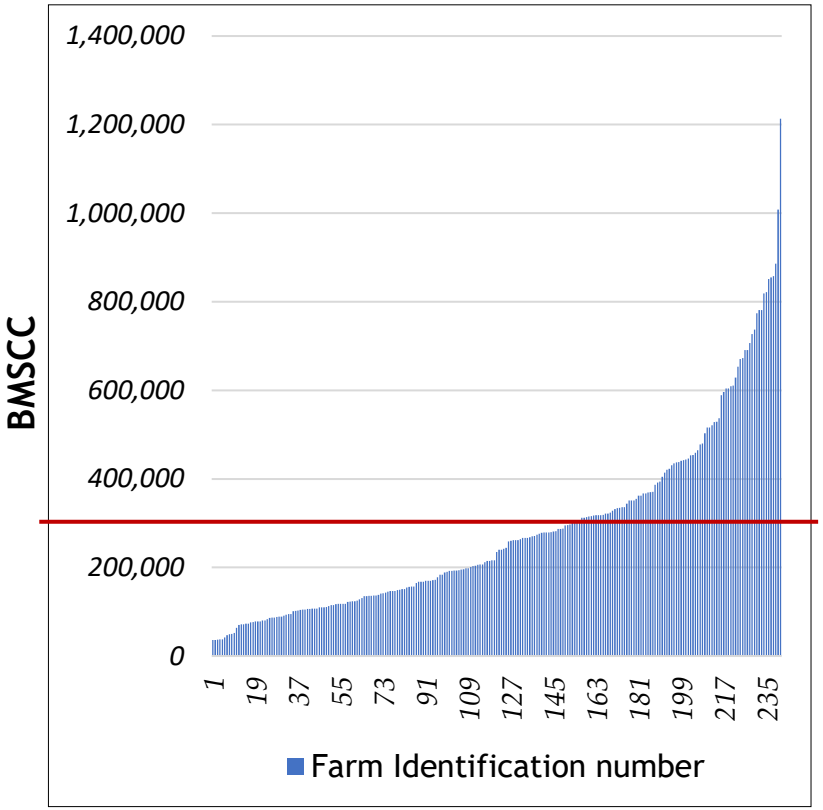
Quarter levels risk factors: Multivariable mixed-effects logistic regression analysis of subclinical mastitis (defined as CMT score ≥2)

Variable name	Categories	Number of observations	OR (95% CI)	P
Rearing system	Free-ranging	641	Reference	0.004
	Household	308	1.8 (0.8 to 4.2)	
	Intensive	**267**	**6.6 (2.2 to 19.6)**	
	Semi-free-ranging	1,461	1.02 (0.6 to 1.8)	
	Semi-intensive	814	1.0 (0.5 to 1.9)	
Quarter position	**Left quarters**	**1,745**	**1.3 (1.1 to 1.6)**	0.006
	Right quarters	1,746	Reference	

Buffalo levels risk factors: Multivariable mixed-effects logistic regression analysis of subclinical mastitis (defined as at least one of the quarters has CMT ≥2)

Variable name	Categories	Number of observations	OR (95% CI)	P
Udder symmetry	Symmetrical	573	Reference	0.0002
	Asymmetrical	**278**	**1.8 (1.3 to 2.4)**	
History of any previous clinical mastitis in last 12 months	No	820	Reference	
	Yes	**31**	**3.1 (1.2 to 8.0)**	0.02
Number of milkers in the herd	Multiple	407	Reference	0.003
	Single	**444**	**1.5 (1.2 to 2.1)**	

Herd-level Bulk milk somatic cell count

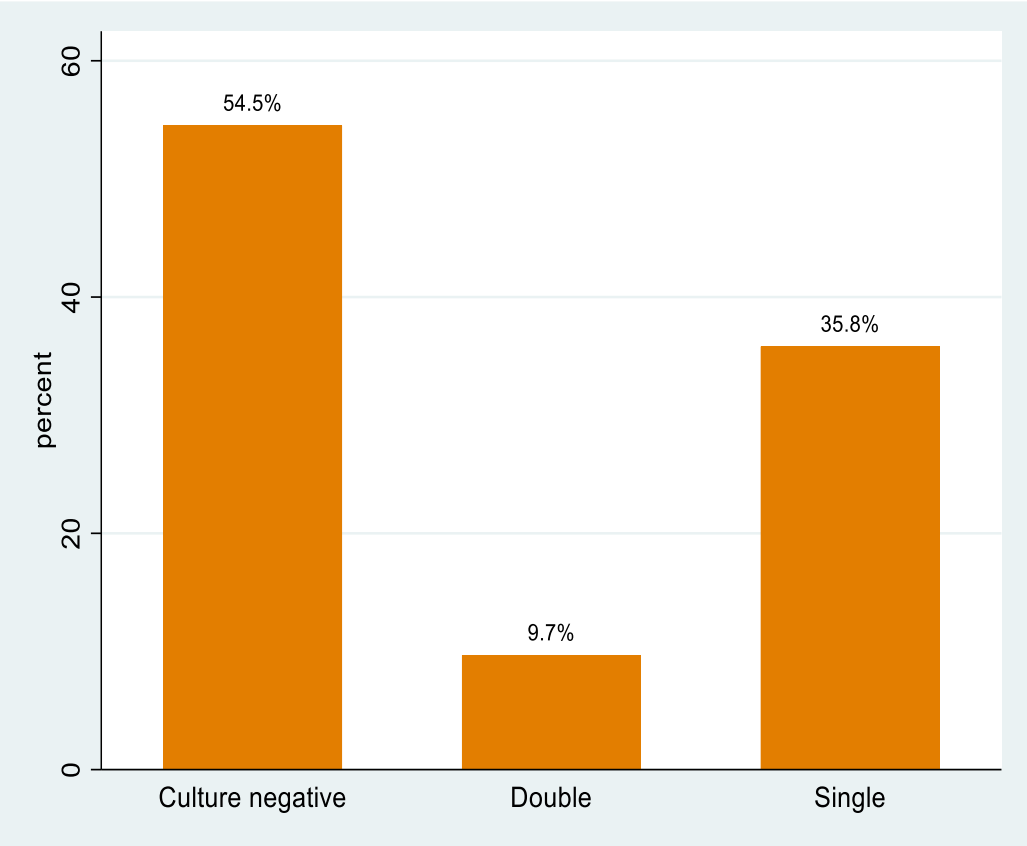


Mean BMSCC: 280,430/mL of milk

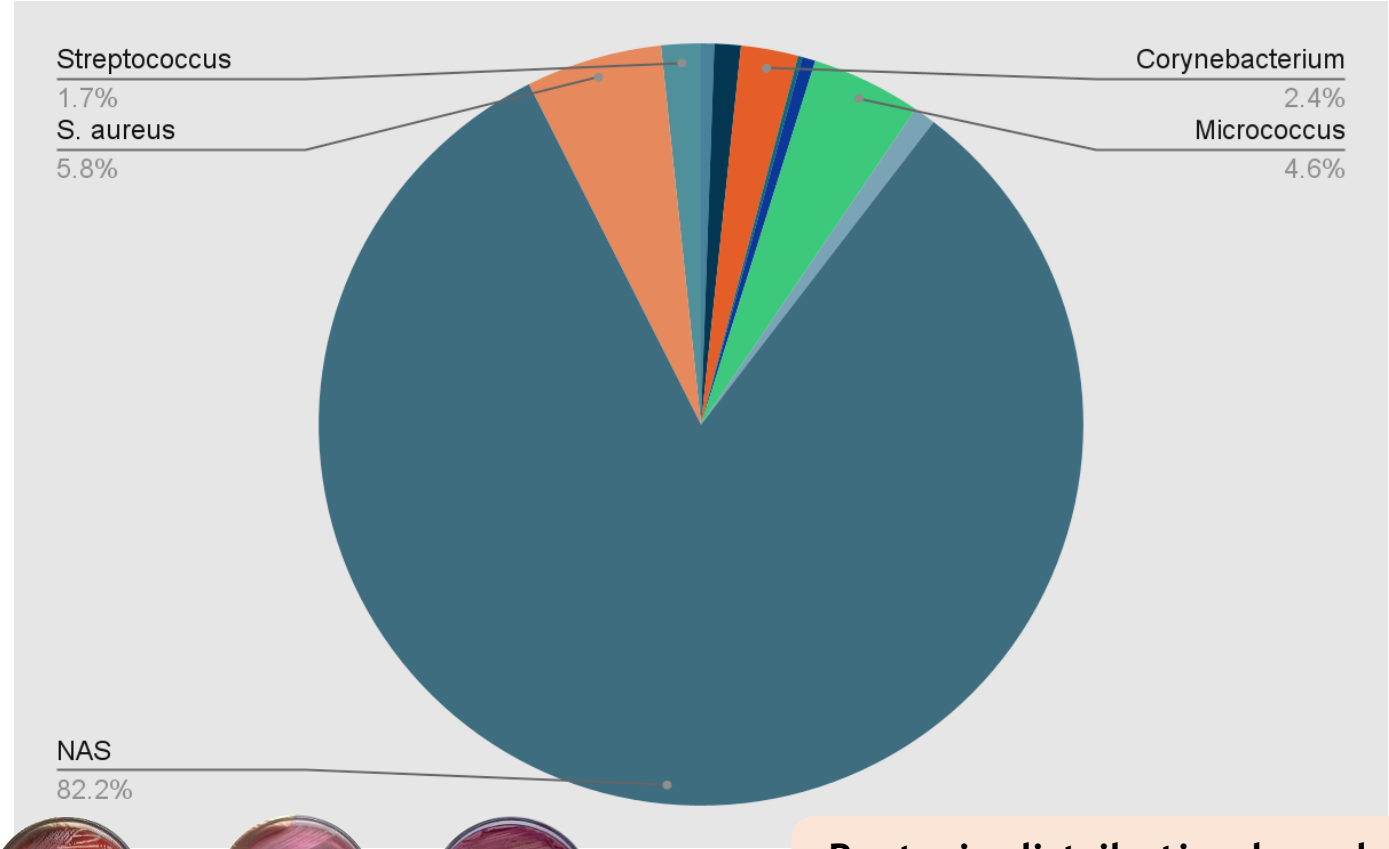
Acceptable BMSCC: 400,000/mL of milk

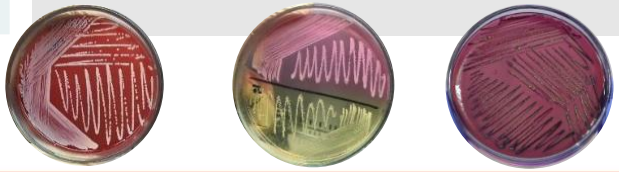
Type of rearing systems	Number of buffalo	Number of herds sampled	Geometric Mean BMSCC (cells/mL)
Free-ranging	**164**	**48**	**209,124**
Household	**77**	**33**	**273,311**
Intensive	**69**	**4**	**412,120**
Semi-free-ranging	**366**	**103**	**200,062**
Semi-intensive	**204**	**54**	**217,231**

IMI Pathogen (N=625) distribution from 1374 buffalo quarter milk samples



IMI pathogens from the collected milk samples





Bacteria distribution based on MALDI TOF MS confirmation

Conclusions

1. **High prevalence** of animal and quarter level subclinical mastitis
   - It reflects mastitis is also an important prevalent disease of buffalo like dairy cows

2. **Mean bulk milk somatic cell count is satisfactory and largely varies between farms**
   - Is an indication of good buffalo milk quality and possibilities to adopt management following better farms

3. **Non-aureus Staphylococci appeared as the most frequent IMI causing pathogens**
   - Needs further assessment of the risk for AMR and virulence characteristics

4. **SCM was most common in intensive farms**
   - So careful assessment of buffalo management is needed in this rearing system in Bangladesh and also applicable to the similar management system in other countries

5. **Left quarters were at higher risk for SCM**
   - Hygienic measures are necessary to prevent IMI thereby reduce SCM

6. Increased risk of SCM observed in **Asymmetrical udders, buffalo with previous history of CM, and single milkers**
   - There is a need of controlled management for these risk group of buffalo population

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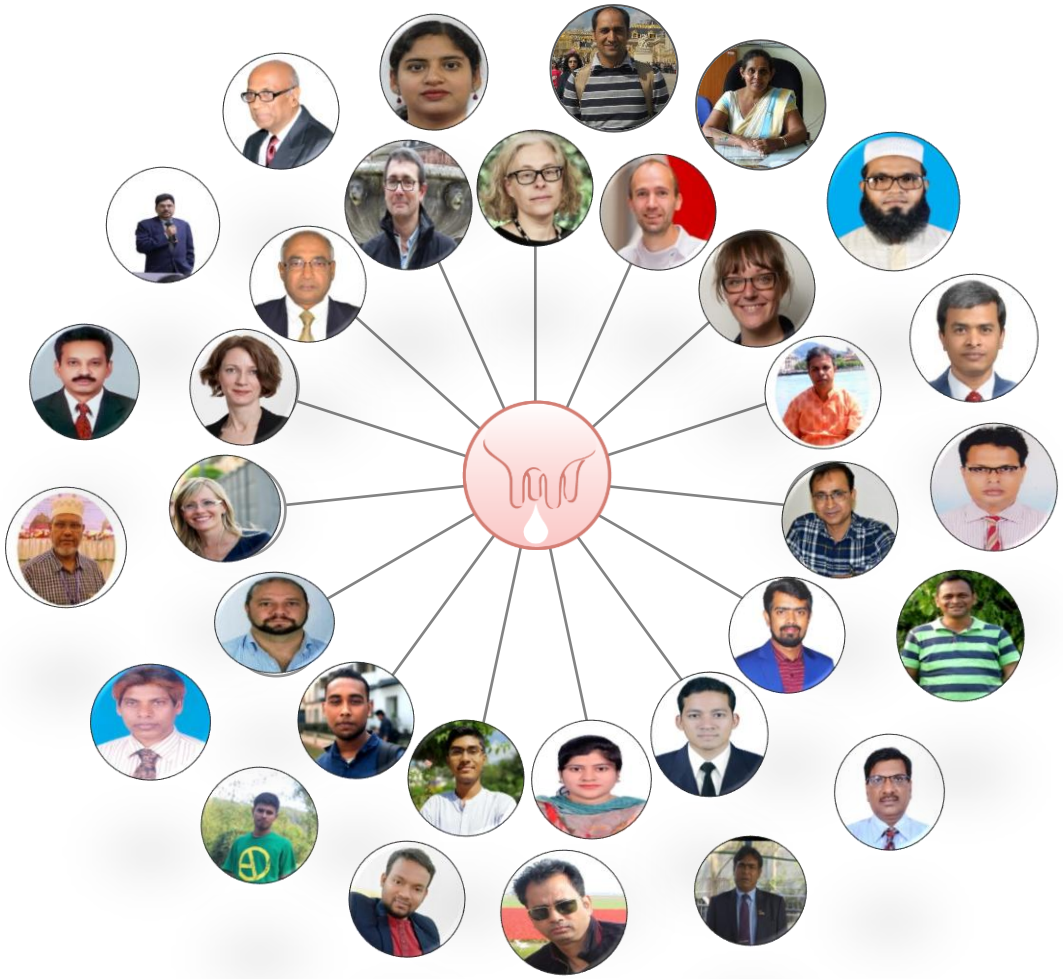
Udder Health Bangladesh Team



Grazie mille!!
ধন্যবাদ!!